Purification of nucleic acids

An important step in many molecular biology and analytical chemistry experiments is to isolate, purify, or concentrate nucleic acids from aqueous solutions that contain undesired proteins, lipids, sugars, or salts.

The precipitation of nucleic acids with ethanol or isopropyl alcohol as antisolvent (*Alternative A*) is an exceptionally quantitative method that recovers even sub-nanogram quantities of precious material in high purity, and picogram quantities when a carrier is added. The precipitate can be dissolved in any volume and buffer of choice.

For higher throughput, nucleic acids can be immobilized on glass particles or silica membranes in the presence of chaotropic agents, such that contaminants can be washed off (*Alternative B*) (Vogelstein and Gillespie, 1979). Binding is quantitative, but elution may be incomplete or volume-restricted. Fragments shorter than 50 bp or longer than 20 kbp are not efficiently recovered, making this method unsuitable for small RNAs such as siRNA, miRNA, or tRNA. For plasmid preparations, where the scale is set by the culture volume and the tip, see SOP 0020.

Neither method discriminates between DNA and RNA. To enrich for one class of nucleic acids, digest with RNase A or DNase I prior to purification.

Risk assessment

– Guanidine salts form TOXIC GASES with bleach or strong acids

▷ Wear gloves, safety glasses, lab coat

☐ DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Oct 27, 2026

Procedures

A › *Purification by alcohol precipitation*

- ☐ 3 M NaOAc, pH 5.2 ◇ R0045
  - *or*: 5 M NaCl ◇ R0046
  - *or*: 10 M NH_4OAc ◇ R0002
  - *or*: 8 M LiCl ◇ R0223
- ☐ 100% Ethanol, ice-cold
- ☐ Buffer PE (Wash buffer) ◇ R0011
- ☐ 5 mM Tris-Cl, pH 8.0 (◇ R0056 for 1 M)
- ☐ 5 g/L LPA (*optional*)

(1.) Add a precipitating salt to the sample at the final concentration listed below; mix well. Compute the spike volume from the available stock.

Sample contains	Salt	Final	Caveat
DNA or RNA (default)	NaOAc	0.3 M	
Long RNA (selective)	LiCl	0.8 M	Wash the pellet well; carry-over may inhibit in vitro translation
dNTPs or oligosaccharides	NH_4OAc	2.0–2.5 M	Not compatible with T4 PNK
SDS	NaCl	0.2 M	

Note: If the sample already contains monovalent cations, less or no additional salt may be required.

(2.) *Optional:* Add 10–20 µg/mL linear polyacrylamide to precipitate picogram quantities of nucleic acids. ⊞

(3.) Add 2.0 vol ethanol to precipitate DNA, or at least 3.0 vol ethanol to precipitate RNA. Mix by vortexing.

(4.) Precipitate for 15–30 min on ice.

Hint: For samples containing less than 0.1 ng/µL DNA or fragments shorter than 100 nt, extend the incubation to 60 min or more and add 10 mM magnesium chloride. Lowering the temperature below 0 °C does not improve recovery: it increases viscosity and slows sedimentation.

(5.) Centrifuge at 12 000 × *g* for 15 min at 4 °C. Discard supernatant and spin briefly to remove residual solvent. The pellet will appear nearly transparent.

Hint: For short fragments or low concentrations, centrifuge for 30 min: time spent spinning recovers more material than time spent incubating.

Hint: Keep the supernatant until recovery has been confirmed for precious samples.

(6.) Wash pellet with 650 µL Buffer PE or freshly prepared 70% ethanol. Discard supernatant carefully.

(7.) Dry pellet for 10–15 min at room temperature, until no ethanol remains but the pellet is still damp.

Critical: Do not dry under vacuum and do not overdry. Desiccated pellets dissolve slowly, and double-stranded fragments shorter than 400 bp may denature.

(8.) Resuspend in 15–30 µL 5 mM Tris pH 8.0 or buffer of choice.

Note: Pellet may adhere to tube walls. To maximize recovery, roll a droplet of buffer along the inner wall several times.

[GS16]

B › *Purification over a silica matrix*

- ☐ Buffer PB (Binding buffer), pH 5.0 ◇ R0010
- ☐ Buffer PE (Wash buffer) ◇ R0011
- ☐ 5 mM Tris-Cl, pH 8.0 (◇ R0056 for 1 M)
- ☐ Spin column, silica, about 25 µg dsDNA capacity (1 / 1 rxn)

(1.) Mix 1.0 vol sample with 2.0 vol Buffer PB.

Hint: To exclude fragments shorter than about 100 bp, dilute Buffer PB with 3–5 vol water before mixing; 1–3 vol suffice for PCR buffers containing detergents or betaine. The cut-off shifts with the fragment and the PCR system, so determine the ratio in advance.

(2.) Apply up to 700 µL to a silica spin column. Centrifuge at 11 000 × *g* for 30 s or use vacuum manifold.

Hint: Silica membranes bind about 25 µg DNA, so it is often possible to reuse the membrane for more of the same DNA. Loading capacity varies by model.

(3.) Wash with 650 µL Buffer PE.

(4.) Repeat wash to minimize salt carryover.

(5.) Dry membrane by centrifugation at 11 000 × *g* for 1 min.

Critical: Residual ethanol may persist if flow-through is not discarded prior to drying.

Critical: Before salt-sensitive applications such as blunt-end ligation, remove the ethanol completely: centrifuge 5 min at 11 000 × *g*, or incubate the open column 2–5 min at 70 °C.

(6.) Place spin column in a clean tube. Apply 15–30 µL 5 mM Tris pH 8.0 onto the silica membrane. Incubate 1 min. Elute at 11 000 × *g* for 1 min.

Hint: For fragments longer than 1 kbp, warm the elution buffer to 70 °C and incubate 5 min on the membrane before spinning.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

Analyses

○ Measure nucleic acid concentration and purity at 260 nm, 280 nm, and 230 nm. The conversion factors below assume a 1 cm path length.

Nucleic acid	A_{260} = 1.0	A_{260}/A_{280}	A_{260}/A_{230}
dsDNA	50 ng/µL	1.8–1.9	2.0–2.2
ssDNA	33 ng/µL	1.8–1.9	2.0–2.2
ssRNA	40 ng/µL	1.8–2.0	2.0–2.2

> ***Note:*** Extinction coefficients and purity ratios vary with sequence. A_{260}/A_{280} shifts with pH by 0.2–0.3 [WMC97].

> ***Quality assurance:*** A low A_{260}/A_{230} indicates residual chaotropic salt, EDTA, carbohydrate, or phenol. Guanidinium thiocyanate absorbs strongly at this wavelength; the guanidine hydrochloride of Buffer PB does not, so a depressed ratio after this procedure points to the other contaminants.

Troubleshooting

Purification by alcohol precipitation

In Step 8:

– Pellet does not dissolve
  - Use low-ionic strength buffer initially; avoid $MgCl_2$ or high-salt buffers.
  - If needed, dissolve in larger volume and repeat precipitation to exchange buffer.

Purification over a silica matrix

In Step 6:

– Little or no DNA in eluate
  - Ensure elution buffer is low salt and within pH 7.0 and pH 8.5.

Materials

Purification by alcohol precipitation

- ☐ **Precipitating salt**
  - **Sodium acetate (NaOAc)**, 3 M, pH 5.2 ⟋ R0045
  - *or* **Sodium chloride (NaCl)**, 5 M ⟋ R0046
  - *or* **Ammonium acetate (NH_4OAc)**, 10 M ⟋ R0002
  - *or* **Lithium chloride (LiCl)**, 8 M ⟋ R0223
- ☐ **Ethanol**, 100%, ice-cold
  CAS 64-17-5
- ☐ **Buffer PE (Wash buffer)** ⟋ R0011
- ☐ **5 mM Tris hydrochloride (Tris-Cl)**, pH 8.0 (⟋ R0056 for 1 M)
- ☐ **Linear polyacrylamide (LPA)**, 5 g/L (*optional*)
  Hint: Inert carrier for precipitating picogram quantities; co-precipitates with the nucleic acid and makes the pellet visible.

Purification over a silica matrix

- ☐ **Buffer PB (Binding buffer)**, pH 5.0 ⟋ R0010
- ☐ **Buffer PE (Wash buffer)** ⟋ R0011
- ☐ **5 mM Tris hydrochloride (Tris-Cl)**, pH 8.0 (⟋ R0056 for 1 M)
- ☐ **Spin column** (1 / 1 rxn), silica, about 25 µg dsDNA capacity
  Hint: QIAGEN's QIAprep spin column fits the same format and was once named here, but it is not a drop-in: its binding buffer is used at 5.0 vol rather than 2.0 vol, its handbook gives no comparable lower fragment-size cut-off, and the size-selection dilution below is validated for Buffer NTI only. Work out the conditions before substituting it.
  - **NucleoSpin Gel and PCR Clean-up column**
    Macherey-Nagel, NucleoSpin Gel and PCR Clean-up, 740609.50

List of references

M.R. Green and J. Sambrook, *Cold Spring Harbor Protoc.* (2016).

M.R. Green and J. Sambrook, *Molecular cloning : a laboratory manual*, 4th ed. (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y, 2012).

B. Vogelstein and D. Gillespie, *Proc. Natl. Acad. Sci. U.S.A.* **76**(2), 615—619 (1979).

W.W. Wilfinger, K. Mackey, and P. Chomczynski, *BioTechniques* **22**(3), 474—481 (1997).

J.A. Zeugin and J.L. Hartley, *Focus* **7**(4), 1–2 (1985).

Change log

2023-06-10 **Benjamin C. Buchmuller** Adaptation as SOP; ethanol precipitation procedure follows Green & Sambrook, "Molecular Cloning: A Laboratory Manual" (CSH Press, 4th ed., 2012) (Green and Sambrook, 2012).

2026-10-27 **Benjamin C. Buchmuller** Corrected the precipitation thresholds, the ethanol volume for RNA, the small-fragment cut-off, the column capacity, the Buffer PE wash, the elution and drying conditions, Buffer PB to pH 5.0 and the 230 nm note; dropped vacuum drying and the mini/midi/maxi yields, which belong to sop_0020. Credits: Zeugin and Hartley (1985) for the centrifugation conditions and Wilfinger et al. (1997) for the pH shift.

From the *Lab Protocols* collection by B. C. Buchmuller and contributors (2026). Licensed under Creative Commons Attribution Share Alike 4.0 International; see https://creativecommons.org/licenses/by-sa/4.0/ for the terms.

For research use. Provided in good faith, without warranty or liability for any use or results. Users are responsible for compliance with local regulations and institutional policies. Where a brand name, manufacturer, or specific source is cited, it is given as an example and does not imply that other suitable equivalents are excluded.

Current when printed. Visit https://benjbuch.github.io/check/ or scan the QR code to check for updates.

061b00f